
The AI171 Crash: A Comprehensive Disaster Analysis

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General Information

The 12th of June 2025 was a black day in the history of the aviation industry. An Air India (AI) Boeing 787-8 Dreamliner bound for London Gatwick tragically crashed shortly after takeoff from Ahmedabad Airport, India. The aircraft had 242 people on board, including staff, and the disaster claimed 241 lives in the air and 19 more on the ground, as the plane crashed into a residential area. This report seeks to provide a comprehensive analysis of this disaster.

The Boeing 787 Dreamliner aircraft departed from Ahmedabad airport at 1:38 PM IST. (Air India, 2025). Less than a minute after take-off, the aircraft lost contact with Air Traffic Control. Several eye-witness testimonies, CCTV footage of the airport, as well as videos taken by nearby residents, depict the aircraft beginning to descend rapidly, losing altitude. The flight crashed at 0809 UTC immediately, a few minutes after take-off. The aircraft crashed into the residential area nearby.

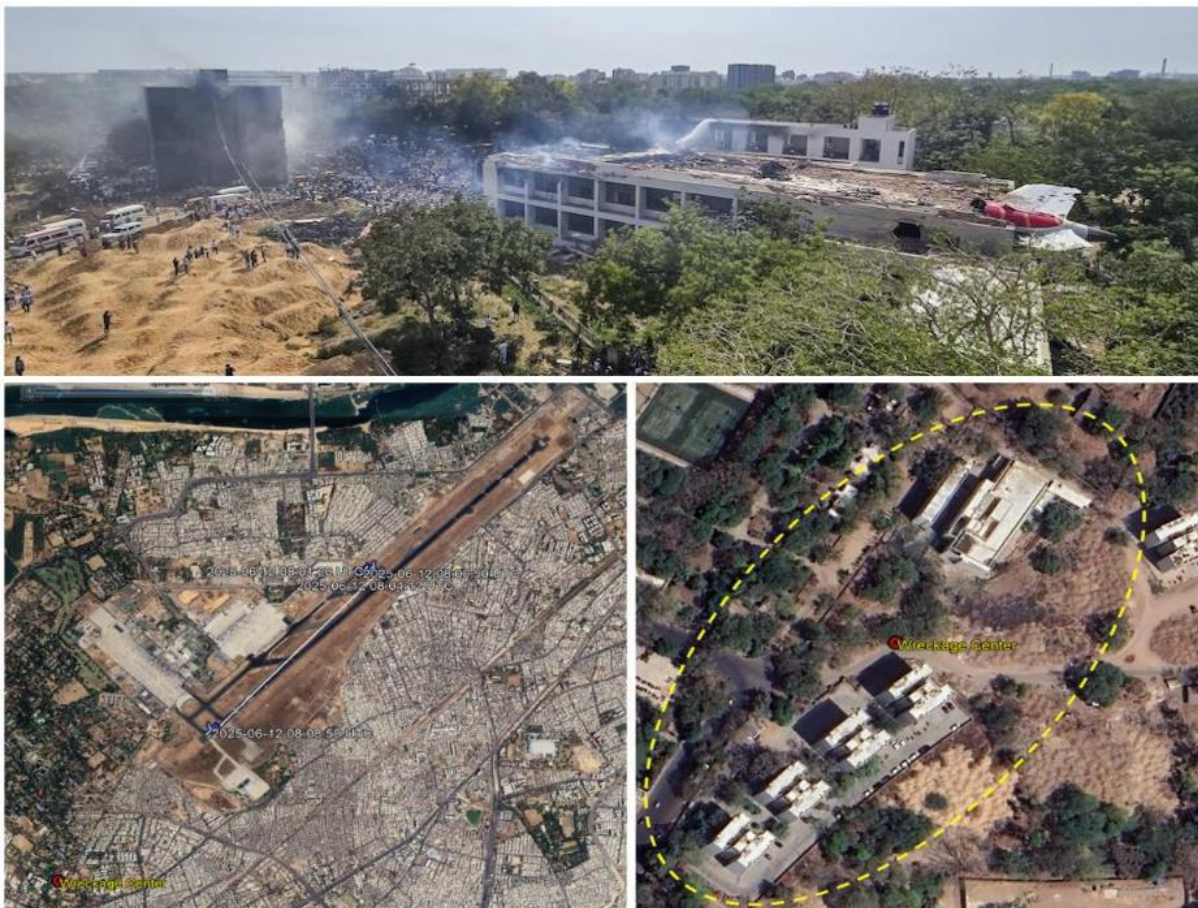


Figure 1 Crash site with Airport and debris field (Aircraft Investigation Bureau, 2025)

The crash resulted in 260 deaths, 241 from the aircraft and 19 on the ground. The crash had only a sole miracle survivor, who is a British national of Indian origin and was seated in seat 11A (Air India, 2025).

Disaster Typology and Hazard Profile

The Air India plane crash can be classified primarily as a technological disaster. The disaster was a result of the failure of man-made technology, equipment and systems. Beyond this, the plane crash can also be classified as a compound disaster. The impact was not confined to the aircraft. The plane crashed in a well-populated residential area. Moreover, as the aircraft was losing altitude, it initially made contact with a series of trees and an incineration chimney inside the Army Medical Corps compound. Parts of the plane struck the campus of BJ Medical College and the city civil hospital in the Meghaninagar locality, destroying a hostel building and igniting fires. Several people on the ground were killed or injured as the wreckage and flames spread through the complex. A total of five buildings shown in the figure below were impacted and suffered major structural and fire damage. The wreckage from the first impact point to the last identified aircraft item was distributed in an area of approximately. 1000 ft * 400 ft (Aircraft Investigation Bureau, 2025).

Time (UTC)	Event
07:43:00	The aircraft requested pushback and startup.
07:43:13	ATC approved pushback.
07:46:59	ATC-approved start-up.
07:49:12	ATC queried whether the aircraft required full length of the runway; the aircraft confirmed requirement of full length of Runway 23.
07:55:15	The aircraft requested taxi clearance, which was granted by ATC.
08:02:03	The aircraft was transferred from Ground to Tower Control.
08:03:45	The aircraft was instructed to line up on Runway 23.
08:07:33	The aircraft was cleared for take-off from Runway 23; wind 240°/06 kts.
08:09:05	MAYDAY call was made by AI171.

Table 1 Sequence of Events (Aircraft Investigation Bureau, 2025)

The accident occurred during the initial climb of the aircraft, where, after only 30-40 seconds of remaining airborne, the aircraft crashed. Flight data shows that the aircraft reached an altitude of about 625 ft above mean sea level and a speed of 180 knots. Immediately after, Engine 1 and Engine 2 fuel cutoff switches transitioned from RUN to CUTOFF position one after another with a time gap of 01 sec, and the flight began descending while still in a nose-up attitude, indicating catastrophic loss of thrust with little height or time for recovery. (Aircraft Investigation Bureau, 2025)

This type of disaster is extremely rare as the aviation industry maintains the most stringent safety measures globally. According to International Air Transport Association (IATA) data, the global commercial jet accident rate is approximately one accident per 1-2 million flights,

the fatal accident rate is approximately 1 per 5-8 million flights, and the dual engine failure rate is extraordinarily rare, estimated at 1 per tens of millions of flights. Accidental fuel cut-off, that is, a complete system failure leading to fuel to both engines getting cut off, is one of the rarest reasons for the failure of an aircraft. Modern aircraft have sophisticated systems and backup controls specifically to prevent this scenario. In fact, this may be the first time that the movement of both fuel control switches to CUTOFF caused dual-engine failure in flight.

Indicators and Data Sources

Indicator	Unit	Value	Year(s)	Data Source
Mortality (Onboard)	Deaths	241	2025	(Aircraft Investigation Bureau, 2025)
Mortality (Ground)	Deaths	19-39	2025	(Aircraft Investigation Bureau, 2025)
Sole Survivor	Individuals	1	2025	(Aircraft Investigation Bureau, 2025)
Seriously Injured	Individuals	67+	2025	(Aircraft Investigation Bureau, 2025)
Total Economic Loss (Insurance Claims)	USD Millions	475	2025	(Fortune India, 2025)
Hull and Engine Loss	USD Millions	125	2025	(Fortune India, 2025)
Liability Claims	USD Millions	350	2025	(Fortune India, 2025)
Ex-Gratia Payment Fund	INR Crores	500	2025	(Air India, 2025)
Interim Compensation per Family	INR Lakhs	25	2025	(Air India, 2025)
Infrastructure Damage	Buildings	3+	2025	(Aircraft Investigation Bureau, 2025)
Time to Crash from Takeoff	Seconds	32-36	2025	(Aircraft Investigation Bureau, 2025)
Maximum Altitude Achieved	Feet	625	2025	(Dr. Ahmed ElHamahmy, 2025)

Table 2 Indicators of Air India's B787-8

Causal Analysis (Hazard–Exposure–Vulnerability Framework)

The official crash investigation started immediately after the crash. The Aircraft Investigation Bureau secured the site, studied the wreckage and tried to get the recordings from the flight. Since the crash occurred in Indian airspace, the investigation will also happen under the control of state regulators, manufacturers of the aircraft, and regulators from around the world. All the remains of the aircraft were secured and collected to rebuild the aircraft, along with documentation. As per regulations, the investigations start almost immediately, and the report is required to be published within 30 days. A simulation exercise is carried out to assess what could have gone wrong.

For the AI 171 disaster, a High-Level Multi-Disciplinary Committee was constituted to examine the causes leading to the crash. The Committee is also expected to review the current SOP (Standard Operating Procedures) and suggest policy and operational reforms to manage such incidents.

The Committee included the following:

- Union Home Secretary, Ministry of Home Affairs, Government of India (Chairman).
- Secretary, Ministry of Civil Aviation, Government of India
- Additional Secretary/ Joint Secretary, MHA
- Representative from the State Home Department, Government of Gujarat
- Representative from State Disaster Response Authority, Government of Gujarat
- Police Commissioner, Ahmedabad
- Director General, Inspection and Safety, Indian Air Force
- Director General, Bureau of Civil Aviation Security/ DG BCAS
- Director General, Directorate General of Civil Aviation/DG DGCA
- Special Director, IB
- Director, Directorate of Forensic Science Services, Government of India (Aircraft Investigation Bureau, 2025).

The committee was chaired by the Union Home Secretary and included representatives from the Ministry of Civil Aviation, the Ministry of Home Affairs, the State Government of Gujarat,

and the Directorate General of Civil Aviation (DGCA). The goal was to conduct a high-level investigation, emergency safety measures for the Boeing fleet, black box recovery, and its analysis.

The official cause for the crash has not been declared yet, but there are several hypotheses about the cause of the crash based on the preliminary report released by the Aircraft Investigation Bureau.

It was identified from the preliminary investigation that the fuel control switches had been cut off. Usually, even if this happens, technically, there are multiple indicators that will send warnings well in time to take action. The probability of a software issue or glitch was not ruled out. Moreover, the preliminary report revealed that recovered voice recordings from the aircraft showed one of the pilots asking the other why did he cutoff the engines and the other pilot responding that he had not done so (Aircraft Investigation Bureau, 2025).

The causal analysis of the disaster indicates that the hazard became severe because the aircraft had a heavy fuel load for the international route. The flame had a temperature of 1500 degrees Celsius (2700 degrees Fahrenheit), which further hindered rescue operations post-crash.

Hazard Characteristics

The primary hazard for the crash was a complete loss of engine power. The preliminary report mentions the deployment of the Ram Air Turbine, which is a lesser-known safety measure in an aircraft. The Ram Air Turbine (RAT) was deployed during the initial climb, immediately after take-off. (Aircraft Investigation Bureau, 2025). The RAT is only deployed in cases of extreme emergencies when there is a total engine or electrical failure, to generate emergency power for essential flight controls and instruments. (Air India Crash: Declassified, 2025).

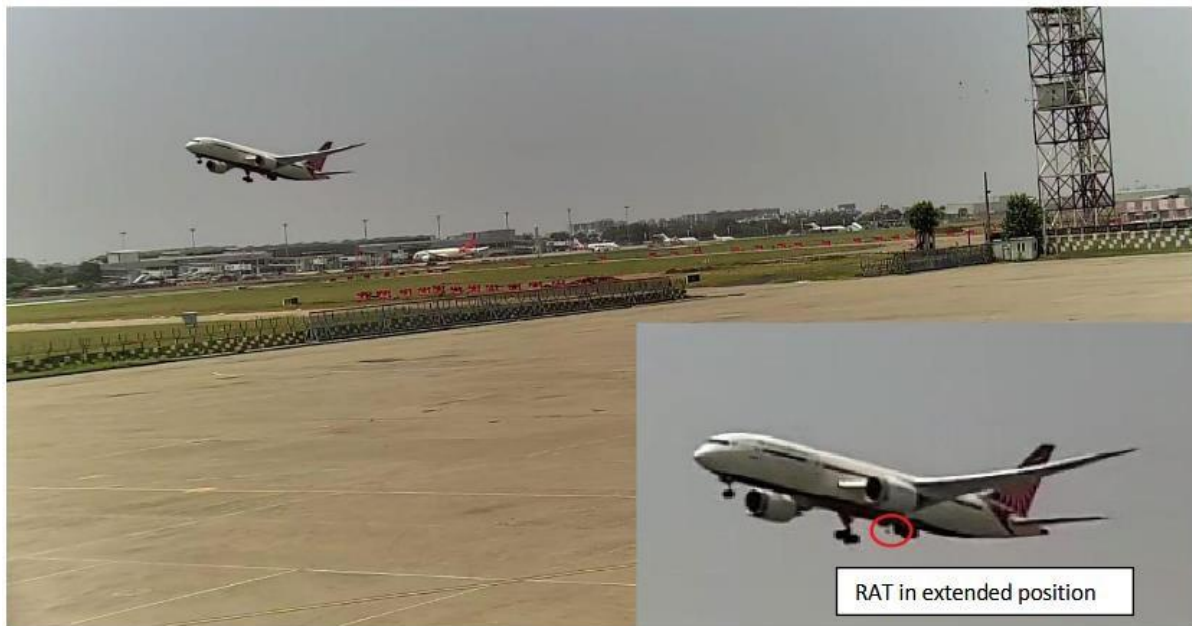


Figure 3 CCTV footage shot of the RAT in extended position (Aircraft Investigation Bureau, 2025)

There were several technical anomalies that may also have contributed to the disaster. According to the preliminary report, in 2018, the Federal Aviation Administration (FAA) issued Special Airworthiness Information Bulletin (SAIB) No. NM-18-33, which spoke about the potential, inadvertent disengagement of the fuel control switch locking feature. The SAIB was issued after it was reported that in model 737 aircrafts, these fuel control switches were installed with the locking feature disengaged. However, this was not considered unsafe, and the recommended changes were not implemented as they were advisory and not mandatory.

The severity of the hazard also increased because the aircraft had a heavy fuel load for the international route. The flame had a temperature of 1500 degrees Celsius (2700 degrees Fahrenheit), which further hindered rescue operations after the crash.

Exposure

All 242 individuals aboard the aircraft were the primary exposed, with no way of escape and no mitigation measures. Secondary exposure involved the ground population. The aircraft crashed into a residential area, and ground level exposure included medical students, hospital staff and patients, nearby residents, and informal workers such as tea vendors and street sellers.

The crash occurred in a residential area with 19 on the ground at the B J Medical Hostel. This incident has reignited debates over the need for having airports located far from densely

populated urban areas. A 2022 study named, ‘You’re Surrounded! Measuring the Enclosure of Airports has ranked the world’s top 50 airports most affected by urbanisation, and Ahmedabad is included in the list. Geographers Tais Grippa and Frédéric Dobruszkes prepared the report based on the population density within a 15-kilometre radius of the airports. The higher the density around an airport, the greater the risk of casualties in the event of a plane crash. Residents living near airports are also exposed to higher levels of noise and air pollution.

Urban planning standards suggest that a densely populated zone should not exist within a 20-kilometre radius of airports. According to the former general manager of Fire Services at the Airports Authority of India, rapid development around Ahmedabad has pushed aside the norm of maintaining a 3-kilometre buffer zone around the airport. Gujarat Samachar 26th Dec 2025

Ahmedabad’s population was just around 3.1 lakh when the airport was built on what was then the city’s outskirts. Today, Ahmedabad’s population has surged to nearly 93 lakh, and the once-remote airport is now surrounded by thick urban development.

Vulnerability

Urban and geographic factors have introduced several spatial vulnerabilities in this context. Poor planning has placed vulnerable infrastructure like hospitals and colleges directly beneath flight paths. Social and institutional vulnerabilities have also increased the risk. Failure of regulatory bodies to consider aviation safety measures while providing approval to urban expansion has allowed buffer zone standards to become weak. Gaps in coordination, communication, and other factors may have allowed legacy infrastructure to operate in an environment for which it was no longer suitable.

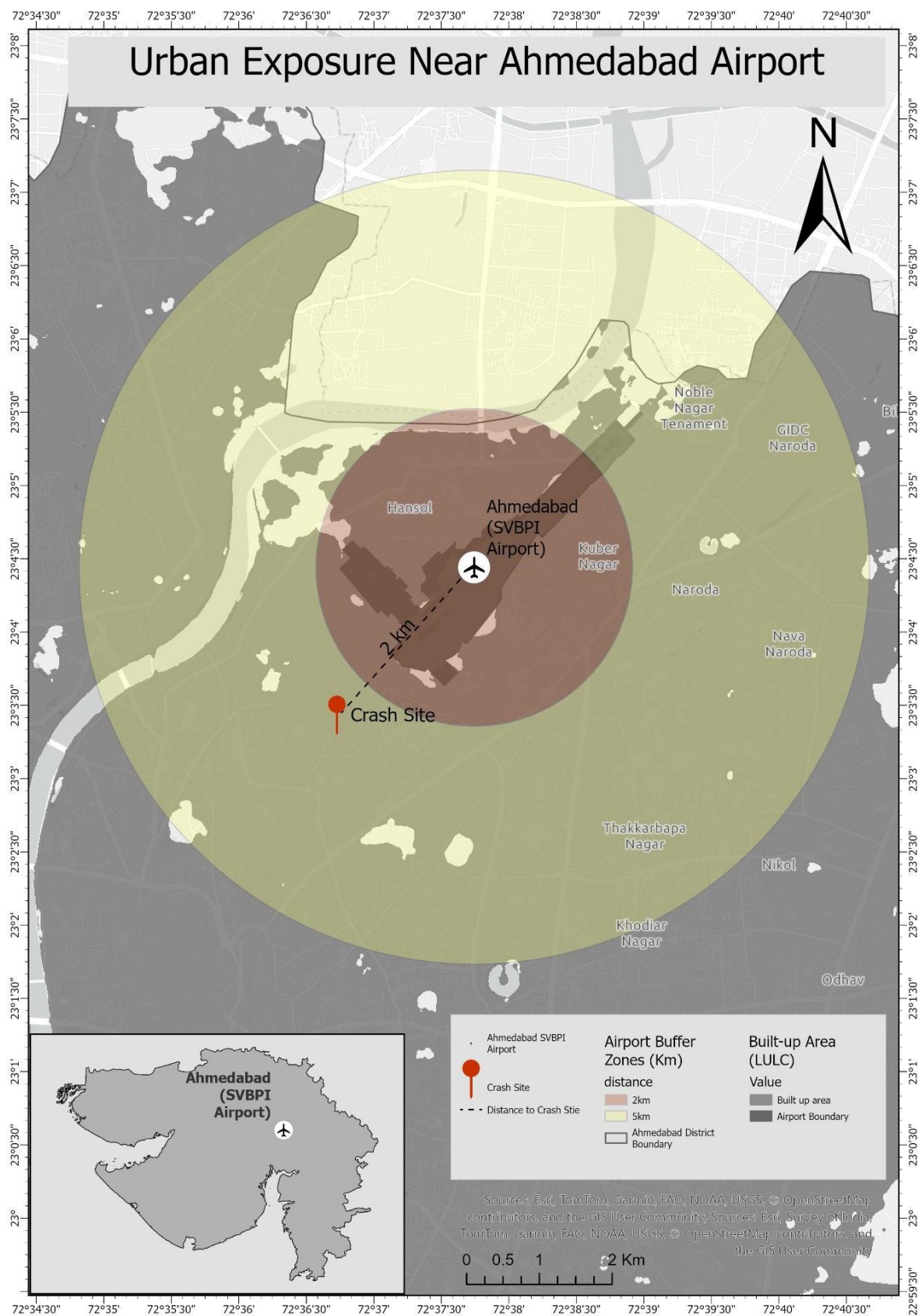


Figure 4 Urban exposure and built-up area around Ahmedabad airport

Impact Assessment

The crash of this aircraft has shaken the aviation industry, as well as the entire country, to the core. It has had a widespread impact across various spheres.

1. Social

Foreign nationals from the UK, Portugal, and Canada, who lost loved ones, as well as Indian families, faced traumatic loss and the long process of DNA identification and repatriation of remains. (Molyneaux, 2025). For the locals, who lived close to the airport, this was a traumatic incident that created a psychological shock in them. Social media spread a lot of misinformation, causing some money racketing activities. (Times Of India, 2025). The loss they faced due to property destruction has yet to be recovered. (Naaz, 2025). Moreover, the livelihood of workers in the BJ Medical College was affected by the loss of lives. Compensation for loss has been promised to all impacted.

2. Economic

The monetary impact of the crash on Air India was significant- loss of aircraft, business loss due to cancellations, insurance claims, compensation to families, investigation costs, etc. The aftermath of a crash triggered a hike in the insurance premiums, which increases the operating cost. The sudden crash resulted in the airport remaining closed for almost 3-4 days, resulting in economic loss. Also, following the crash, the Indian civil aviation regulator, the Directorate General of Civil Aviation (DGCA), announced enhanced safety inspections of Air India's fleet of 33 Boeing 787 Dreamliners. As a consequence, and due to aircrafts being grounded for extensive checks, Air India announced the cancellation of 15 per cent of its wide-body long-haul flights. Governments' costs for investigation, safety audits, support to victims' families, and trauma counselling programs for affected employees are also incurred.

3. Industrial & Safety Impact

The failure of both engines due to fuel cut off resulted in loss of thrust after take off. Ideally, the pilots should have received a prior signal or warning. This raised questions on the routine safety and compliance checks carried out as a standard practice (Times of India, 2025). Today, cockpit ergonomics is becoming increasingly important to avoid any inadvertent change to the safety settings and automatic protection of the aircraft. Transparency of AAIB preliminary reports caused loss of confidence amongst crew members, questioning the investigation standards.

The humongous insurance claims for the passengers, aircraft and third-party damage as per the Montreal convention were pegged at \$300-\$600 million, causing insurers to up their premiums. (Rai, 2025).

4. *Environmental Impact*

The environmental damage caused by this disaster has not been given much attention and has remained largely unaddressed, even in response efforts and official reports. The aircraft, which was loaded with 70,000 – 90,000 litres of fuel for its international route, released approximately 180 -230 tonnes of CO₂ upon impact, and the flames from the crash reached a temperature of 1500 degrees Celsius. (Aircraft Investigation Bureau, 2025). The combustion of jet fuel in the crash would have released toxic fumes and pollutants into the atmosphere, including carbon monoxide, particulate matter, volatile organic compounds, nitrogen oxides, and carcinogenic polycyclic aromatic hydrocarbons, which would have severely degraded the local air quality.

5. *Psychological and Social Impacts*

The intense destruction caused tremendous psychological damage to the viewers and the ground staff. Anxiety and stress among pilots and crew, trying to cope with the loss of colleagues, spiked their absence from work. As per guidelines from DCGA, Air India and IndiGo conducted post-trauma workshops and mental health support for their staff. (Times of India, 2025).

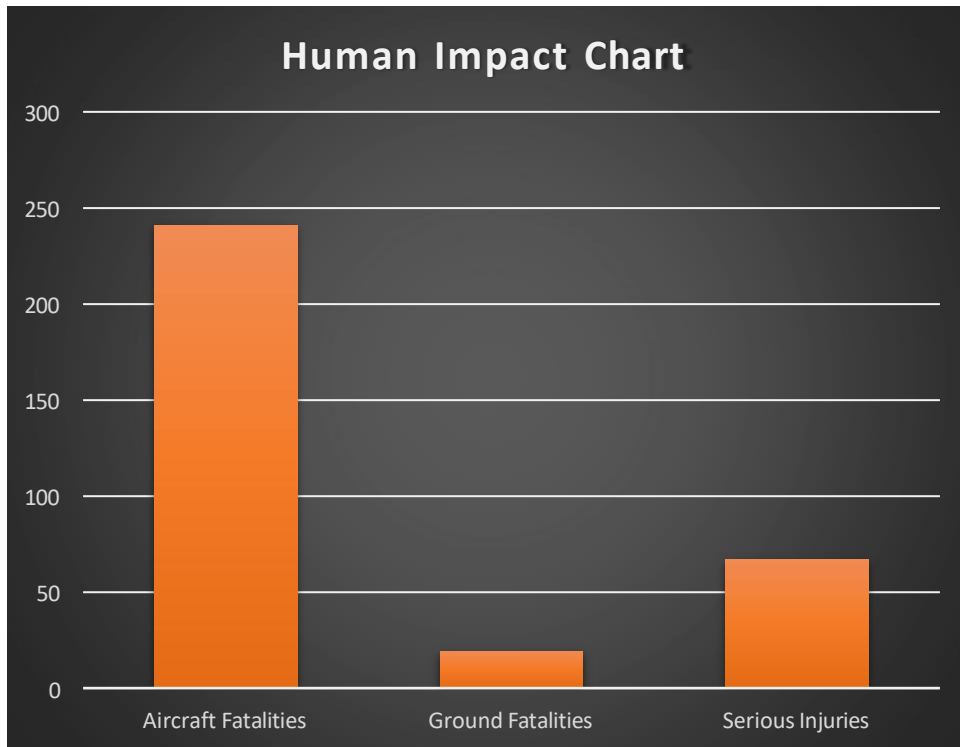


Figure 5 Human Impact Chart

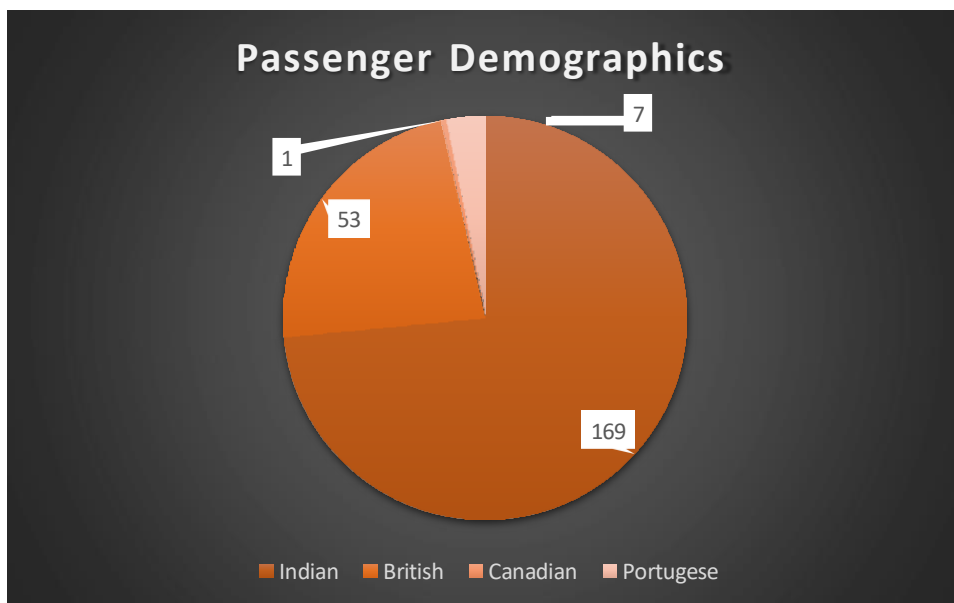


Figure 6 Passenger Demographics (Air India, 2025)

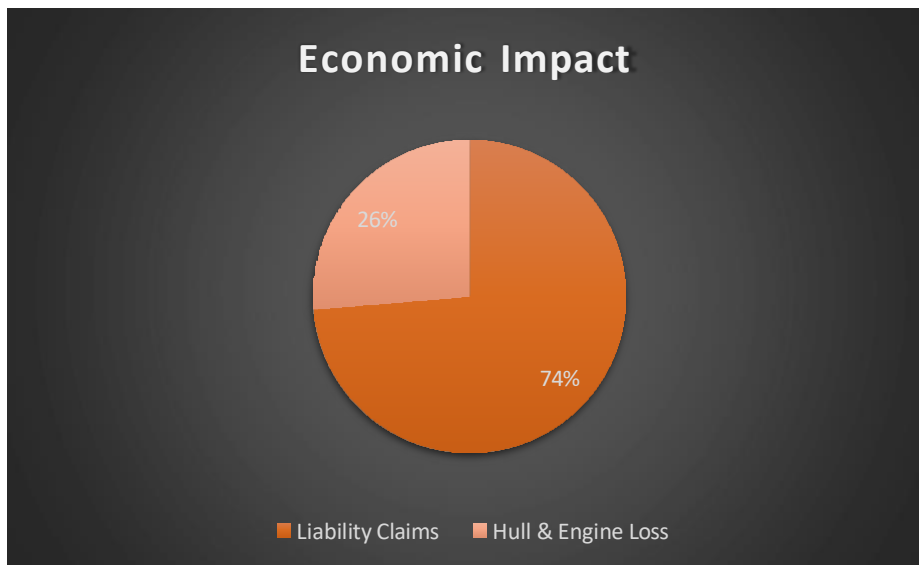


Figure 7 Economic Impact

Application of the Disaster Management Framework

The disaster management framework within the “golden hour”, which is the first 60 minutes, is as follows:

- Confirmation by the Indian Government through a press release, outlining that AI171, a B787-8 Dreamliner, had crashed soon after take-off and that an AAIB investigation was being launched.
- Air India and authorities issued emergency helpline numbers, including a dedicated passenger hotline (1800-5691-444) and separate lines for foreign nationals, to centralise information requests (Air India, 2025).
- The emergency response centre had been activated, and families were being assisted by teams who were giving one-on-one support.
- People living close to the crash site, especially residents near the BJ Medical College area, were among the first to reach the scene, even before some emergency services arrived. They ran toward the wreckage to help pull out injured individuals and check on victims, showing quick courage in the chaotic moments after the crash. (The Economic Times, 2025)
- Chairman N. Chandrasekaran issued an early statement emphasising that the airline’s “primary focus is on supporting all affected people and their families” and on assisting emergency responders at the crash site.

Once the crash site was stabilised, the crisis evolved into two parallel tracks:

1. **Technical investigation** – a domain largely owned by AAIB, Boeing, GE, DGCA, NTSB and FAA.
2. **Humanitarian assistance** – where Air India, Tata Group, and local/state authorities had the primary role.

The severity of the impact and fire caused many bodies to be severely mutilated, complicating identification. Authorities embarked on a demanding DNA-mapping exercise, which was completed by late June.

Within about 10 days, Air India announced a package of ₹25 lakh interim compensation for the families of each deceased passenger and the sole survivor. (Air India, 2025). This was in addition to the ₹1 crore support per family previously announced by Tata Sons. In addition, over 200 trained personnel, including therapists, doctors, nurses and pharmacists for trauma counselling, were deployed in Ahmedabad. All ongoing health needs were also being addressed. A helpdesk to assist with documentation and claims processing was set up. Air India's dedicated AI171 update page and press releases repeatedly emphasised solidarity with families and warned against misinformation. (Air India, 2025).

Tata Sons, within a month of the crash, formalised and completed the registration of a public charitable trust in Mumbai. The Trust will be called 'The AI-171 Memorial and Welfare Trust', dedicated to the victims of the unfortunate accident of Air India flight AI-171 in Ahmedabad.

The Trust will also provide aid and assistance for the alleviation of any trauma or distress suffered by the first responders, medical and disaster relief professionals, social workers, and governmental staff who provided invaluable institutional support and service in the aftermath of the accident.

Every affected family in Ahmedabad was assigned at least one caregiver by Air India. Air India, with support from other Tata group companies, provided every possible assistance to the families and loved ones of those deceased.

India's Aircraft Accident Investigation Bureau was involved in extracting information from the flight data recorder about the engine and control settings, and the voice recorder for cockpit conversations. The Aircraft Accident Investigation Bureau (AAIB) is working to determine the probable cause(s)/contributory factor(s) of the crash.

A team of four officials from the UK's Air Accidents Investigation Branch (AAIB), with expertise in aircraft operations, engineering and flight data, worked in Ahmedabad alongside American and Indian counterparts. Their role was to provide additional support and expertise to the safety investigation being led by India's Aircraft Accident Investigation Bureau. The inquiry was being conducted under international norms as laid out in ICAO Annex 13 and India's Aircraft Investigation of Accidents and Incidents Rules, 2017. It included experts from the Indian Air Force, Hindustan Aeronautics Limited (HAL), and the U.S.-based National Transportation Safety Board (NTSB), officials from Boeing, GE, aviation medicine specialists, and Air Traffic Control experts (CNBC TV 18, 2025)

Evaluation of Disaster Management Effectiveness

Strengths in the response:

- **Helplines and support:** Acknowledging the severity of the disaster, Air India and the government moved quickly in the first hours (Air India, 2025). Emergency centres to deal with interim compensation, medical support, counselling, and individual attention to families of those grieving were all put into action. Air India leadership intervened to express their empathy.

Missed opportunities:

- **Technical communication gap:** The AAIB preliminary report needs to share plain-language explanations for people to comprehend the causality, and continuous briefings will further douse any rumours.
- **Identification of Body:** The sensitivity and the extent of damage due to the explosion required the communication to be extremely careful. A mismatch of body parts leading to a mismatch of DNA was reported after private DNA checks by a British family (The Guardian, 2025)
- **Slow updates and lack of information:** The last update on the website of Air India regarding the crash was on July 26th, 2026 (Air India, 2025). Since then, there have been no updates from either Air India or the government as to what could have been the potential cause of this disaster. Slow updates and lack of information ...

The table below summarises these strengths and weaknesses in a concise manner.

Planned measures	Outcomes
Response time	Rapid support for AAIB, Government, NGO
Fire Control Measures	Rapid but the magnanimity slowed operations.
Medical care	Immediate
Community support	Strong
Interagency coordination	effective
Trauma management	Limited support for long-term trauma
Comprehensive emergency planning	Weak for the surrounding areas outside the perimeter of the airport
Livelihood restoration	Poor
Environment mitigation	Poor

Table 3 Evaluation of Disaster Management Effectiveness

Risk Reduction and Control Measures

The AI171 crisis offers several hard-earned lessons for India and for any country whose airlines operate under global scrutiny:

1. Crisis communication is not a 72-hour exercise.
It is a multi-month, sometimes multi-year discipline. The communication investment must increase when technical reports, legal suits, and cross-border issues surface.
2. Investigators and PR teams must collaborate while respecting independence.
The right of the victim's family to know what the cause was should be respected. Technical findings and any information must be explained in a way that is understood.
3. Digital crisis hubs to disseminate information are now critical. This can be the means for families or next of kin to get updates on the situation and compensation.
4. Structural reforms must be communicable, not just real. Fleet inspections, training overhauls, maintenance audits, all of these must be visible and narrated to regain trust.

5. Scrutiny of existing procedures, certification processes, flight protocols, and safety checks globally becomes critical. The recommendations from the FAA, like the one about the locking mechanism in the fuel control switches, must be carefully deliberated upon before either accepting or rejecting them.
6. Enhanced training, support with AI for decision making in pressure situations, getting a consolidated view of the weather and temperature conditions that may impact aircraft performance, and stricter pre-takeoff checks must be available to the pilot in real time.
7. A collaborative implementation plan involving key aviation stakeholders is crucial to foster compliance and promote a culture of safety. Proactive measures must be pursued as ethical responsibilities, ensuring continued improvement in aviation safety protocols and engaging with international organisations to disseminate best practices globally.

Institutional and Governance Analysis

The AI171 disaster response has revealed both the strengths and the gaps of India's response and governance framework. Investigations into the reason for the crash began promptly, with a High-Level Multidisciplinary Committee being formed to investigate, and the preliminary report was released on time. However, significant weaknesses have also emerged. The failure to maintain the mandatory 3 3-kilometre buffer zone around Ahmedabad Airport (see fig. 4) reflects poor implementation of rules and a lack of regard for the people's safety. The investigation has also been very slow. There have been very few updates in recent times, and today, more than six months after the disaster, there is still no information as to what caused it. These institutional weaknesses underscore the need for stronger regulatory compliance mechanisms and quick and transparent communication systems.

Policy and Planning Recommendations

This section presents a policy-oriented linkage of the Sendai Framework for Disaster Risk Reduction (SFDRR 2015–2030) and the Sustainable Development Goals (SDGs) with reference to the AI-117 aircraft crash, treating it as a technological / transport disaster within an urban-peri-urban setting.

Aspect	Sendai Framework	SDG Link	Actions
Risk Understanding	Priority 1	11	Zone regulations around the airport Protection of vulnerable communities
Governance	Priority 2	16	Governance framework for safety within the airport and perimeter Robust disaster management governance to share the cause and compensate Preparedness and coordination to face disasters
Infrastructure safety	Priority 3	9	Ongoing upgradations of safety norms Use of Artificial Intelligence to predict and prevent failures Review SOPs and trainings for all pilot and maintenance staff at regular intervals
Preparedness & response	Priority 4	3	Trauma centres for post-crash support Long-term mental health support Emergency preparedness Business continuity planning
Livelihood support	Priority 4	8	Rebuild the livelihood of affected people Support the reconstruction of the destroyed area Trust to provide ongoing support for impacted families
Community participation	Priority 4	17	Multi-stakeholder disaster response partnerships Involvement of local communities and NGO's

Table 4 Linking the Air India Crash with the Sendai Framework and Sustainable Development Goals

The devastating crash of Air India Flight AI171 in Ahmedabad has numbed the people and left families mourning. The aviation industry is grappling with an urgent question: How do we predict and prevent such tragedies?

Embracing newer techniques for predictive maintenance, real-time monitoring, and anomaly detection with AI support holds unparalleled potential to reduce the probability of disasters (ePlaneAI, 2025).

Conclusion

This tragic crash stands as a stark reminder of the unpredictability of disasters and human beings' vulnerability to technological disasters when safety systems fail. The widespread destruction caused by this event, 260 lives lost, infrastructure destroyed, and severe social and psychological impacts, this disaster serves as a reminder of the need for a strong and efficient response system, stricter enforcement of regulations and reforms across urban planning, and crisis communication.

As investigations continue, the memory of the victims of the AI171 crash must strengthen the commitment towards newer safety innovations and stricter compliance to ensure that such a tragedy never again happens in Indian skies.

References

- Air India. (2025). *Updates on AI171*. Retrieved from Air India Newsroom: <http://airindia.com/content/air-india/in/en/newsroom/press-release/AI171.html>
- Air India Crash: Declassified* (2025). [Motion Picture].
- Aircraft Investigation Bureau. (2025). *Preliminary Report Accident involving Air India's B787-8 aircraft bearing registration VT-ANB at Ahmedabad on 12 June 2025*. Ministry of Civil Aviation, Government of India.
- Business Today. (2025, July 24). *Air India crash impact: 112 pilots reported sick, govt reveals in Lok Sabha*. Retrieved from Business Today: <https://www.businesstoday.in/latest/corporate/story/air-india-pilots-sick-leaves-rise->

CNBC TV 18. (2025, December 12). *Air India crash six months on: A timeline of the incident and developments so far*. Retrieved from CNBC TV 18: <https://www.cnbctv18.com/india/air-india-crash-six-months-on-a-timeline-of-the-incident-and-developments-so-far-19787230.htm>

Dr. Ahmed ElHamahmy, D. H. (2025). Technical Assessment of the Air India Flight AI171 Crash on June 12, 2025: A Review of Preliminary Data and Contributing Factors.

ePlaneAI. (2025, June 12). *AI in Aviation Safety: Lessons from the Tragic Air India Flight AI171 Crash*. Retrieved from ePlaneAI: <https://www.eplaneai.com/news/ai-and-aviation-safety-insights-from-the-air-india-flight-ai171-crash>

Fortune India. (2025, June 27). *Air India Flight 171 crash may trigger over ₹3,966 cr payout, among biggest in aviation insurance history*. Retrieved from Fortune India: <https://www.fortuneindia.com/business-news/air-india-flight-171-crash-may-trigger-475-million-payout-among-biggest-in-aviation-insurance-history/124102>

Ghosh, S. (2025, June 12). *AI in Aviation Safety: Lessons from the Tragic Air India Flight AI171 Crash*. Retrieved from Analytics Insight: <https://www.analyticsinsight.net/artificial-intelligence/ai-in-aviation-safety-lessons-from-the-tragic-air-india-flight-ai171-crash>

Molyneaux, I. (2025, June 13). *Air India Flight 171: what we know so far about the tragic plane crash*. Retrieved from AeroTime: <https://www.aerotime.aero/articles/air-india-flight-171-what-we-know-so-far-crash>

Naaz, F. (2025, June 13). *'Felt like earthquake, sky turned black': Ahmedabad resident recalls moment when Air India plane crashed*. Retrieved from livemint: <https://www.livemint.com/news/felt-like-earthquake-sky-turned-black-ahmedabad-resident-recalls-moment-when-air-india-plane-crashed-11749798144364.html>

Rai, V. (2025). *Air India Crash in Ahmedabad: Safety, Operational and Regulatory Concerns*. Singapore: Institute of South Asian Studies, National University of Singapore.

- The Economic Times. (2025, July 13). *'Like a loud blast': What eyewitnesses saw as Air India flight AI171 crashed into BJ Medical College dining hall*. Retrieved from The Economic Times: <https://economictimes.indiatimes.com/news/new-updates/like-a-loud-blast-what-eyewitnesses-saw-as-air-india-flight-ai171-crashed-into-bj-medical-college-dining-hall/articleshow/121806304.cms>
- The Guardian. (2025, July 23). *British families 'sent wrong remains' after loved ones killed in Air India crash*. Retrieved from The Guardian: <https://www.theguardian.com/world/2025/jul/23/british-families-sent-wrong-remains-loved-ones-air-india-crash>
- Times of India. (2025, July 7). *Ahmedabad crash fallout: DGCA directs AI & IndiGo to conduct post trauma workshops for 'worried' crew members*. Retrieved from Times of India: <https://timesofindia.indiatimes.com/india/ahmedabad-crash-fallout-dgca-directs-ai-indigo-to-conduct-post-trauma-workshops-for-worried-crew-members/articleshow/122284059.cms>
- Times of India. (2025, July 12). *Air India crash: What AAIB report confirms, what it rules out, what still remains unanswered*. Retrieved from Times of India: <https://timesofindia.indiatimes.com/india/air-india-crash-what-aaib-report-confirms-what-it-rules-out-what-still-remains-unanswered/articleshow/122407735.cms>
- Times Of India. (2025, July 02). *Air India plane crash: AI-generated fake reports, videos spreading misinformation ; fraudsters exploiting vulnerability*. Retrieved from Times of India : <https://timesofindia.indiatimes.com/business/india-business/air-india-plane-crash-ai-generated-fake-reports-videos-spreading-misinformation-fraudsters-exploiting-vulnerability/articleshow/122202668.cms>